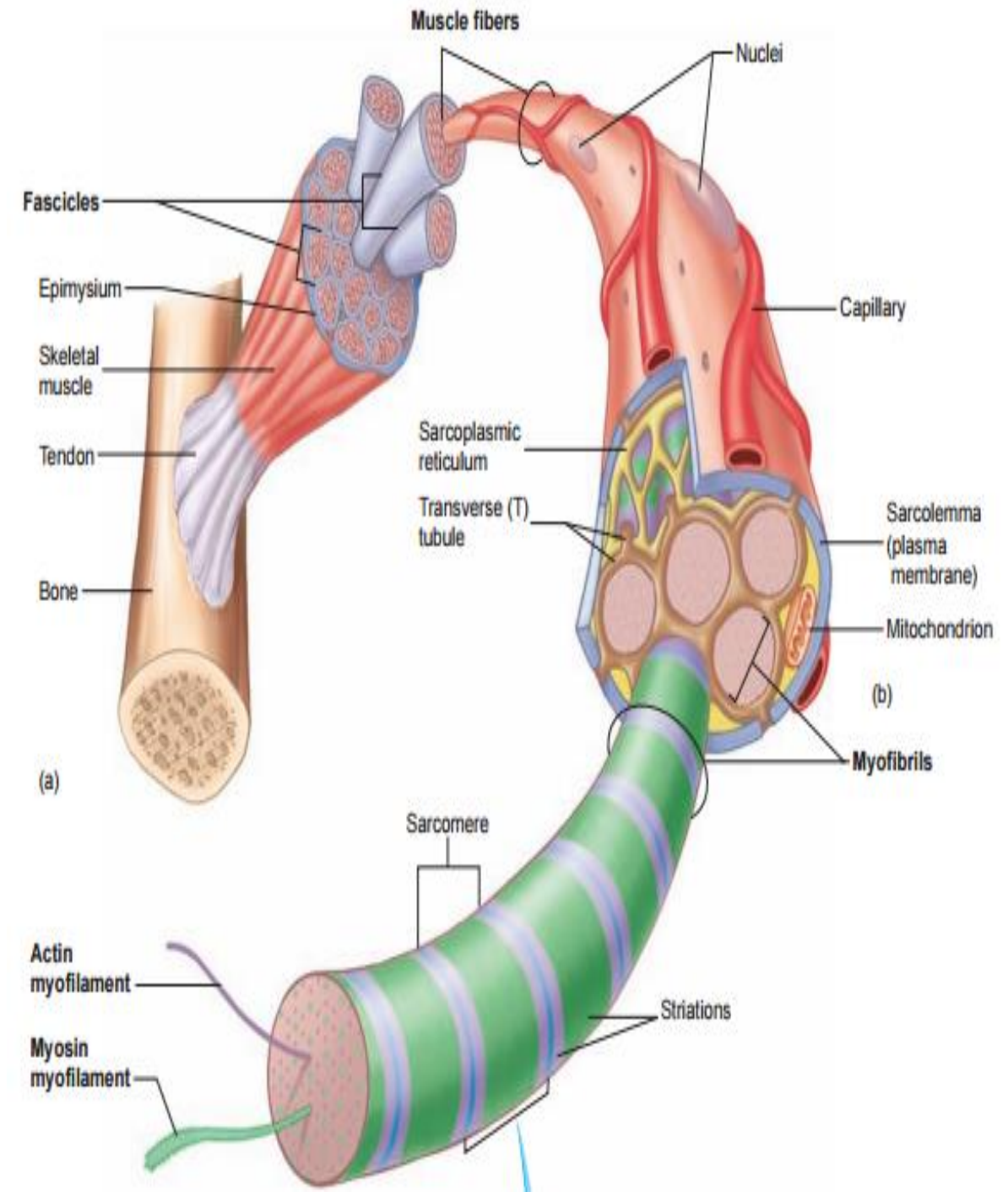


Physiology of Musculoskeletal System



MUSCLE

- ◆ **Muscle** is the organ of the body that converts **potential energy** of food into **mechanical energy**.
 - **Muscle Cells** have a special capacity to produce **force** and **movement** that enable us to speech and to manipulate objects
- ◆ Accounts **40%** of the BW

Types of muscles: 3 types

■ Skeletal muscles

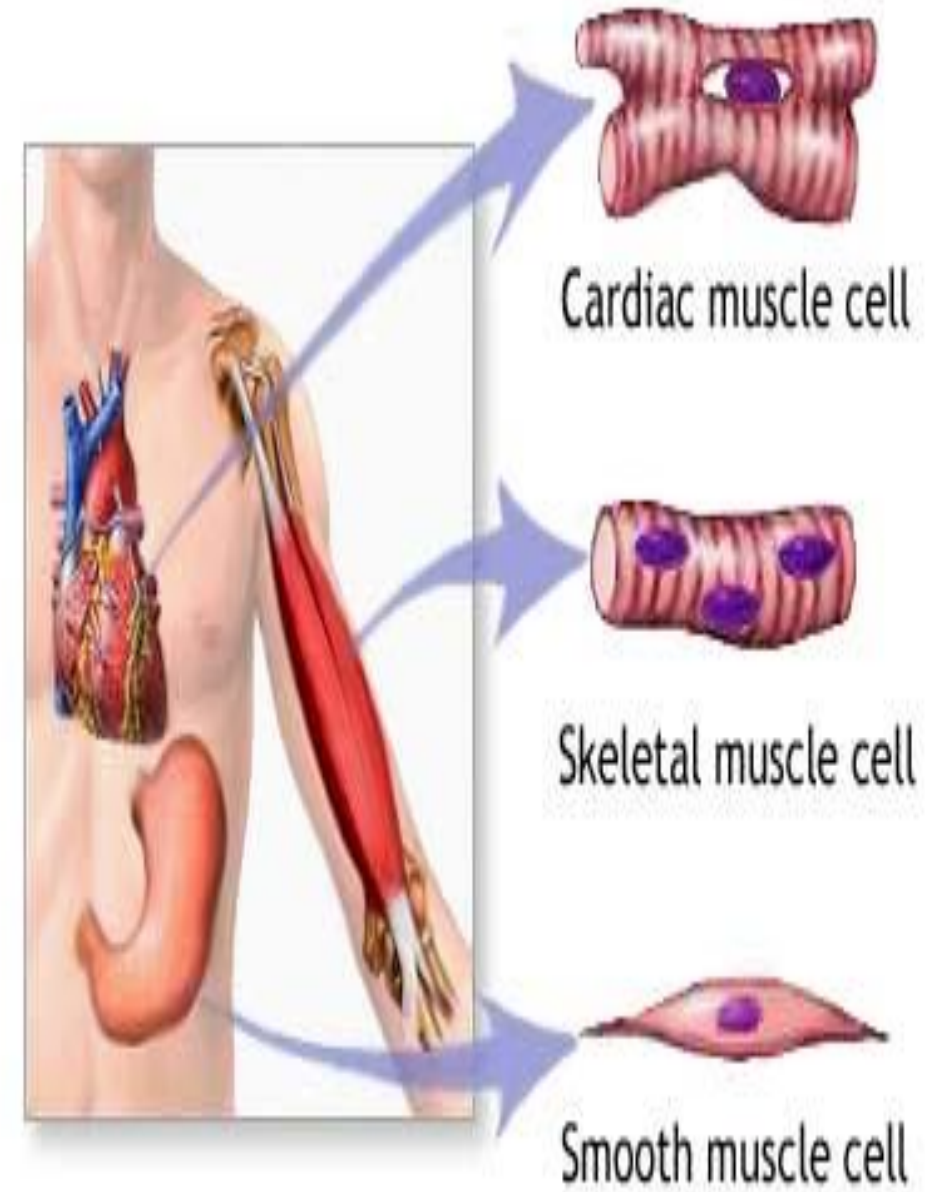
- Located attached to **bones** and moves skeleton
- They are elongated, cylindrical and **multinucleated cells**
- They are **striated muscle**
- They are **voluntary muscle**, controlled by SNS

■ Cardiac Muscle

- muscle of the heart
- They are **striated**
- They are **involuntary**, controlled by ANS, drugs and hormones
- Have the property of **autorhythmicity**

■ Smooth Muscles

- Located in the **wall of hollow internal structures** (GIT, blood vessels, uterus, U-bladder, Iris)
- They have **non-striated appearance, mononucleated cells**
- Involuntary muscle, controlled by ANS, drugs and hormones
- Have the property of **autorhythmicity**



Muscle Functions

1. Movement

- **Body parts**
- **Blood** throughout the body
- **Lymph** through the lymphatic vessels
- **Food** through the GI tract
- **Bile** out of the gallbladder and into the digestive tract
- **Urine** through the urinary tract
- **Semen** through the male and female reproductive tracts
- **A newborn** through the birth canal

2. Stabilizing body positions and posture

- Muscle contraction is constantly allowing us to remain upright.
- The muscles of your neck are keeping your head up right now.
- As you stand, your leg muscles keep you on two feet.

3. Thermogenesis

- **Generation of heat.** Occurs via **shivering** – an involuntary contraction of skeletal muscle.

4. Stabilization of joints

- Muscles keep the tendons that cross the joint nice and firm.
- ▶ This does a wonderful job of maintaining the integrity of the joint.

Characteristics of Muscle Tissue

1. **Excitability** $\Rightarrow$ ability of muscle tissue to receive and respond to a stimulus
 - **Smooth muscle**, the stimulus could be a neurotransmitter, a hormone, stretch, ΔpH , ΔPco_2 / ΔPo_2
 - **Cardiac muscle**, the stimulus could be a NT, a hormone, or stretch
 - **Skeletal muscle**, the stimulus is a NT released by a neuron.
2. **Contractility** $\Rightarrow$ ability to **shorten forcefully** when adequately stimulated.
 - This is the defining property of muscle tissue.
3. **Extensibility** $\Rightarrow$ ability of muscle tissue to **stretch** without damaged
4. **Elasticity** $\Rightarrow$ ability to **recoil** and **resume** to **original length** after stretched or contracted.

Structural Organization of Skeletal Muscle

Entire muscle



Muscle fascicles



Muscle fibers

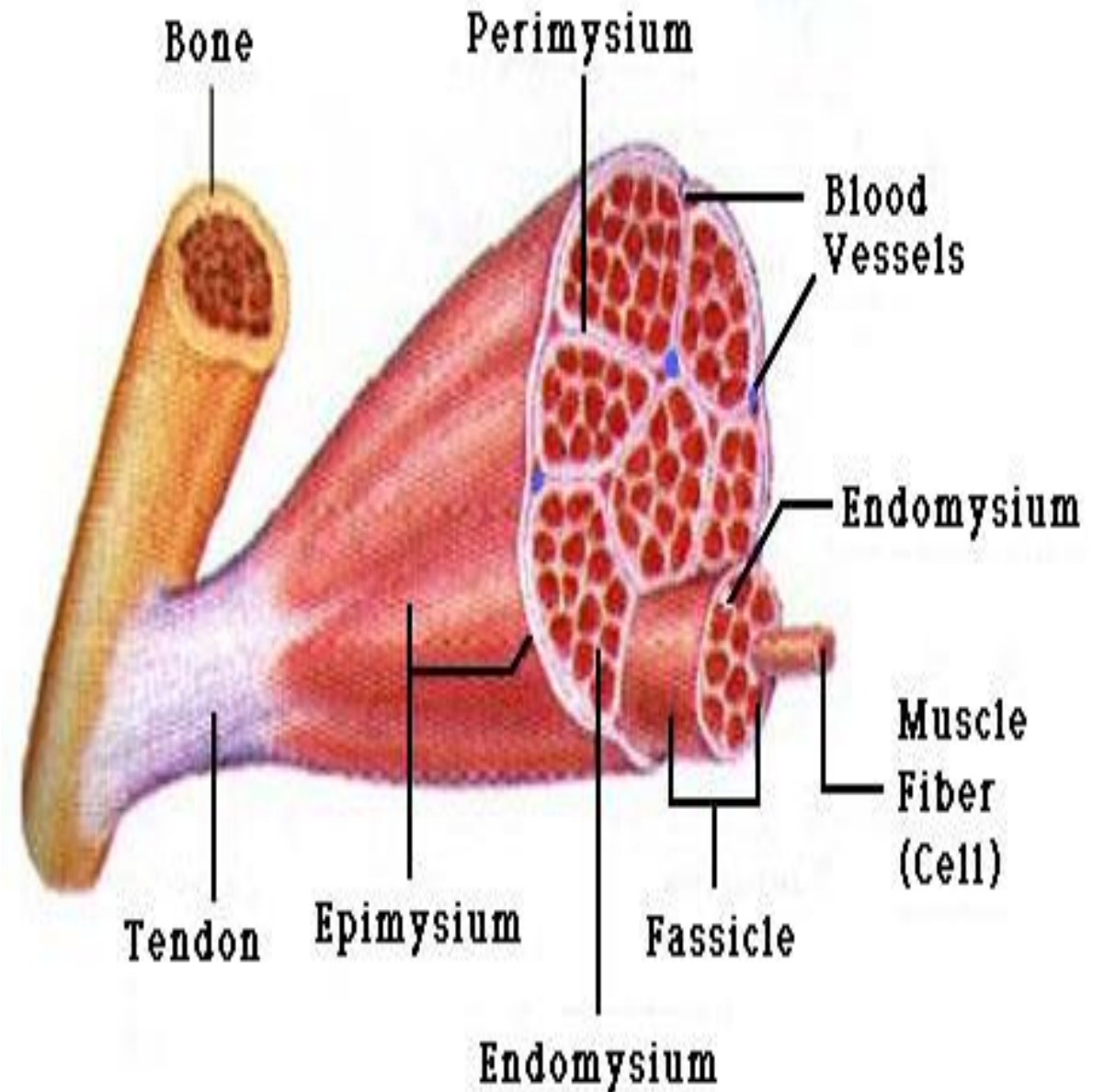


Myofibriles



Myofilaments

- Thin myofilament (actin)
- Thick myofilament (myosin)



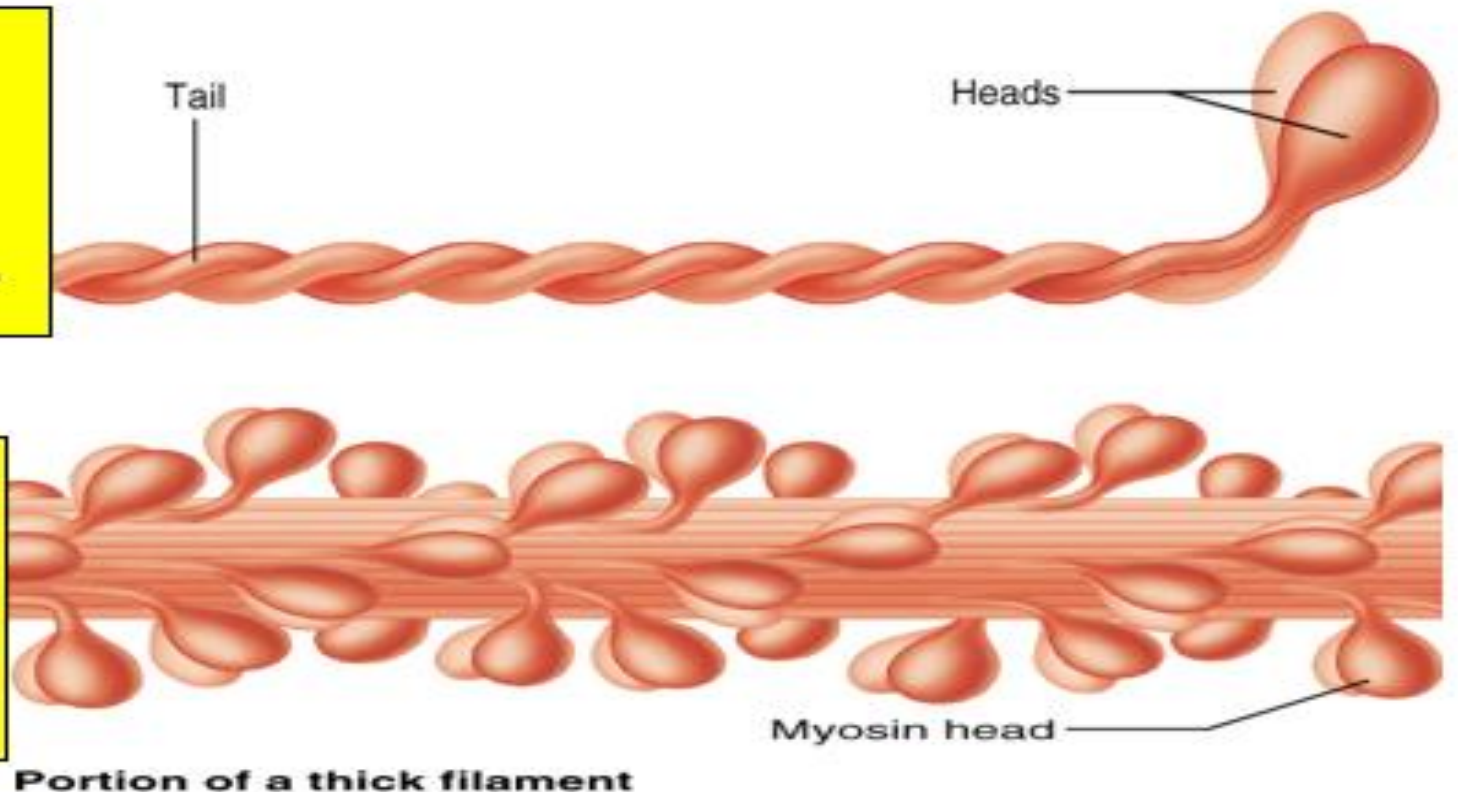
Myofilaments

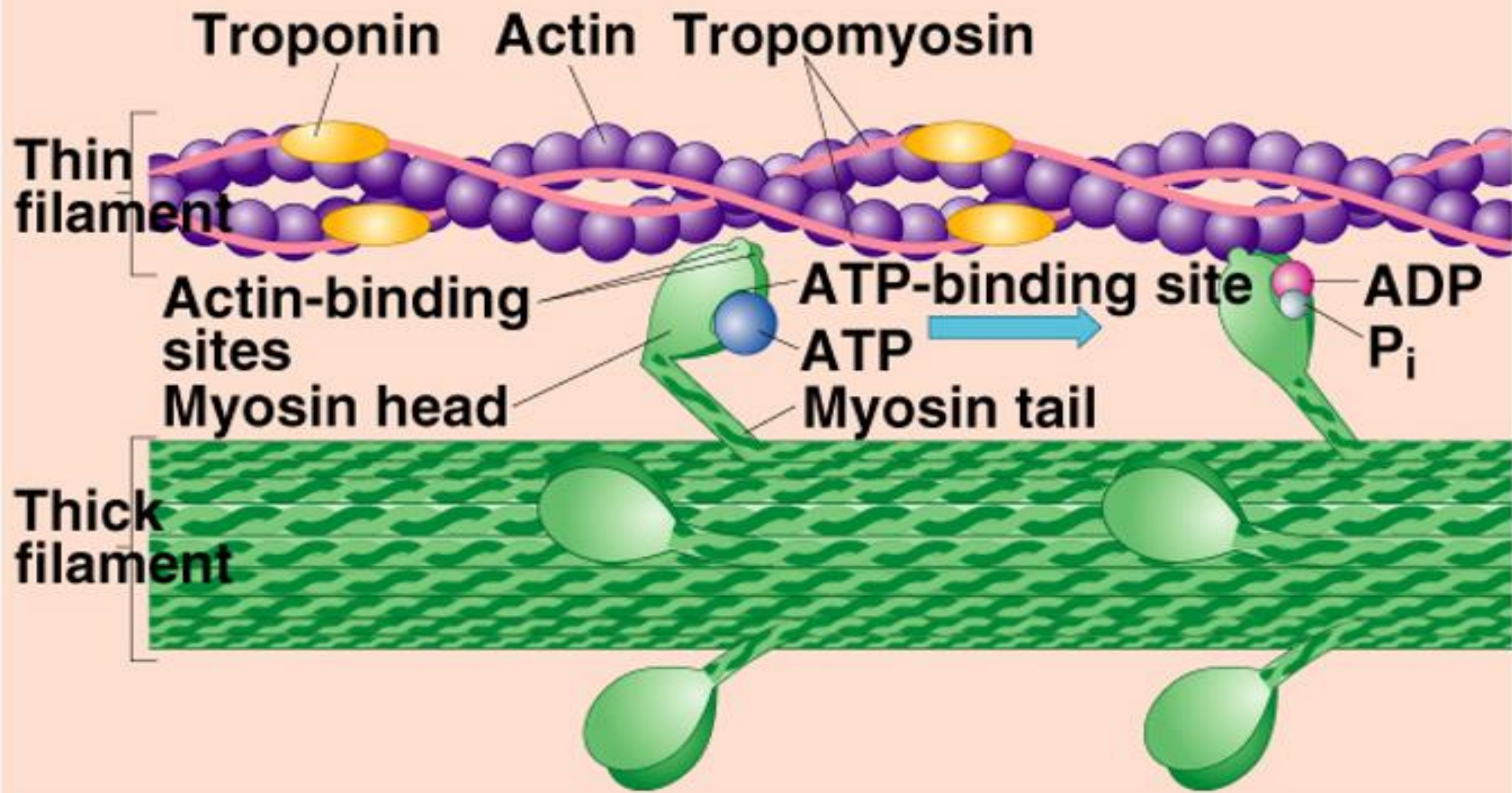
❑ **Two Types** of myofilaments (**thick and thin**) make up myofibrils

■ **Thick Myofilaments** are made up of the protein **myosin**

A single myosin protein resembles **2 golf clubs** whose **shafts** have been **twisted** about **one another**

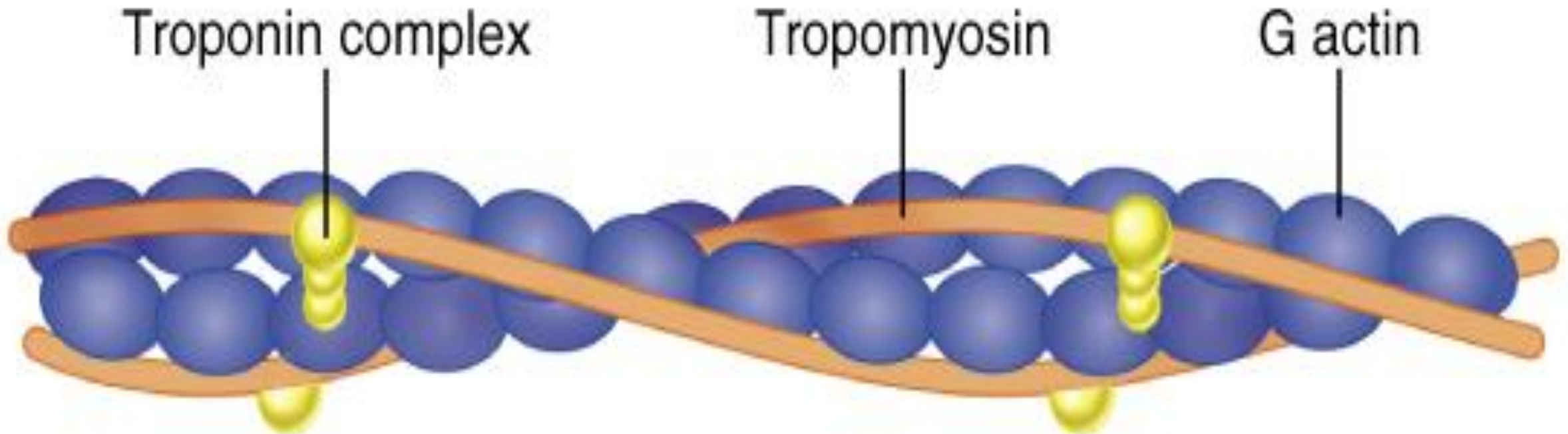
About **300** of these **myosin molecules** are **joined together** to form a **single thick filament**





Thin Myofilament is made up of 3 different types of protein: **Actin**, **Tropomyosin** and **Troponin**.

- ✗ **Actin** $\Rightarrow$ has a binding site for **myosin head**
- ✗ **Tropomyosin** $\Rightarrow$ cover the active site of actin
- ✗ **Troponin** $\Rightarrow$ a trio of proteins (**Troponin-I**, **Troponin-C** & **Troponin-T**) that bound to both actin and tropomyosin



Skeletal Muscle Fiber

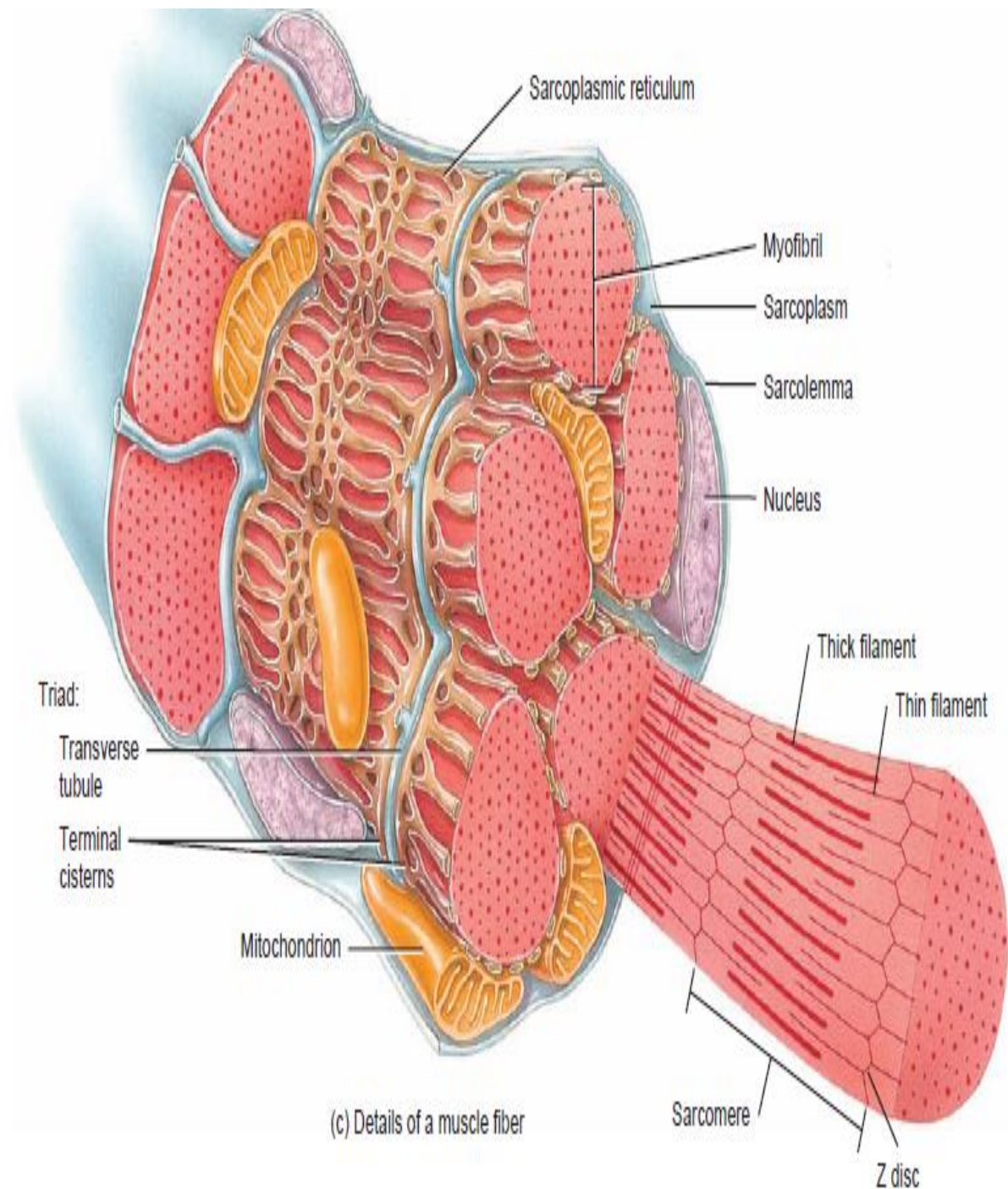
- ❑ Cell membrane is sarcolemma
- ❑ Cytoplasm is sarcoplasm
- ❑ sER is sarcoplasmic reticulum
- ❑ skeletal muscle cell has two tubular structures:

A. Transverse tubules (T-tubules)

Function: Conduction of AP

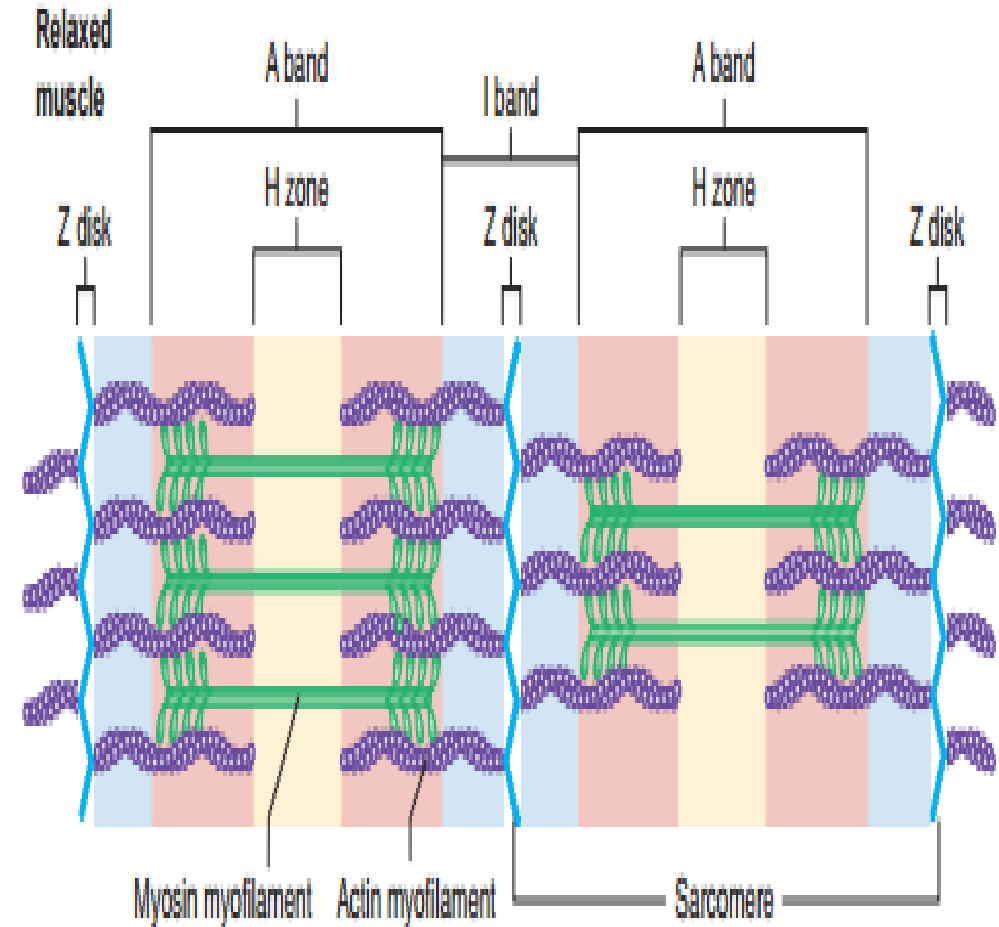
B. Longitudinal tubules (sER)

Function: Ca^{2+} storage



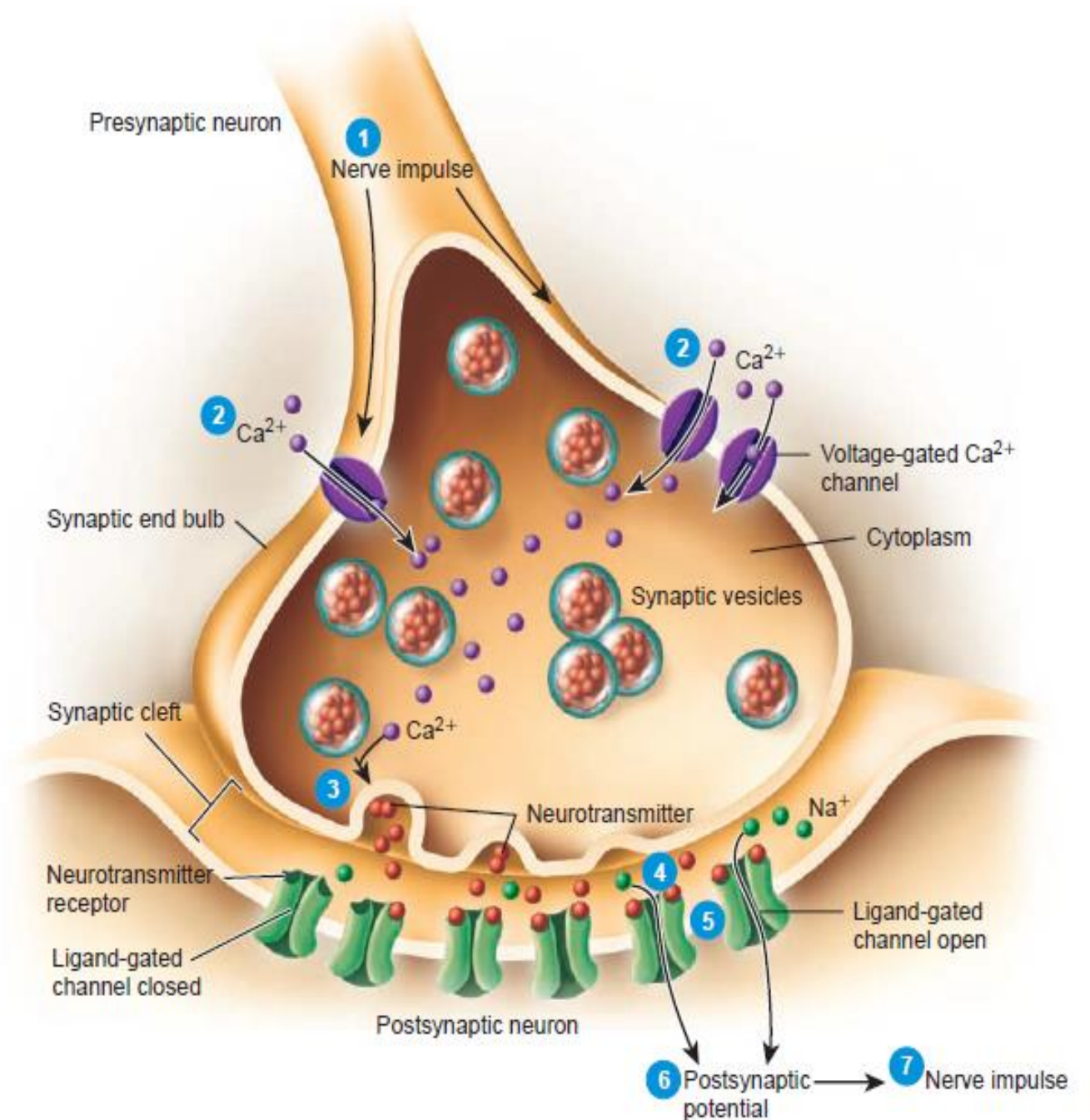
Sliding-Filament Mechanism / Model

- ❑ During muscle contraction, myosin heads of the thick filaments **pull on** the thin filaments, causing:
 - Thin filaments slide toward the center of sarcomere
 - As the thin filaments slide, the **I bands** and **H zones** become narrower and eventually disappear when the muscle is maximally contracted
- ❑ **Sliding-filament mechanism of muscle contraction**, occurs **only** when **high level of calcium ions** and **ATP** is available



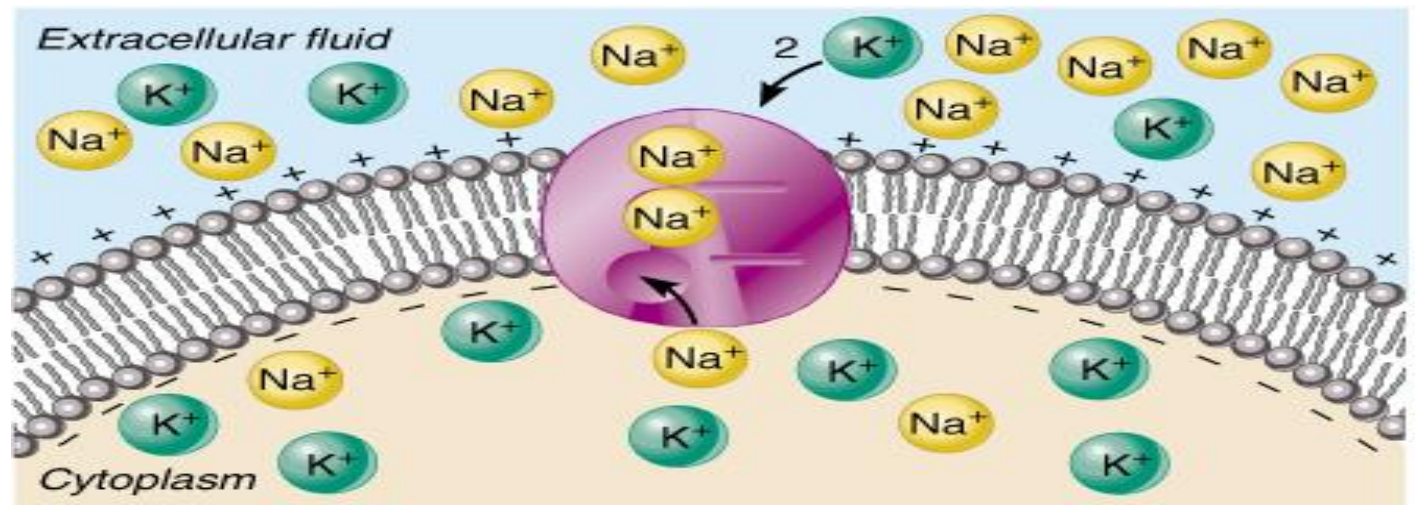
□ The whole process of muscle contraction has 4 steps:

- 1) Excitation
- 2) Excitation-contraction coupling
- 3) Contraction
- 4) Relaxation



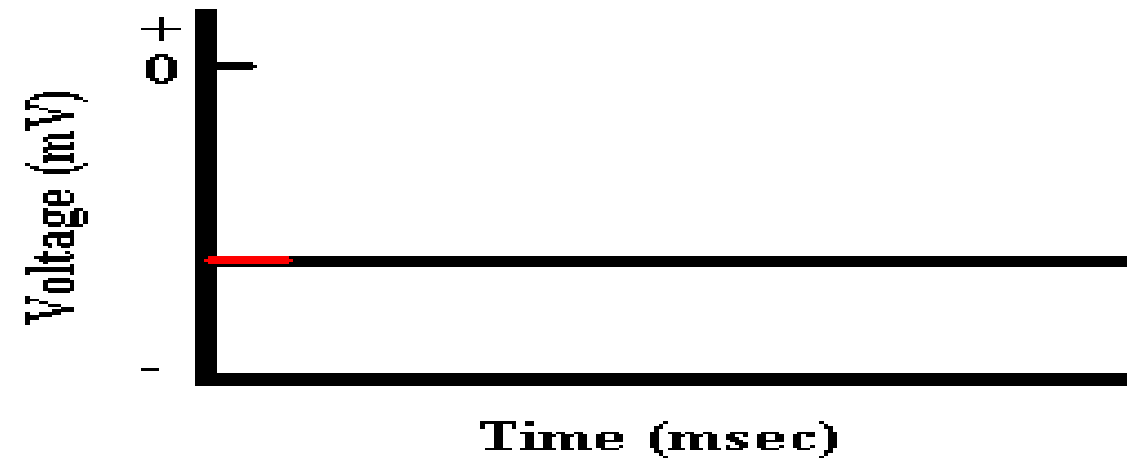
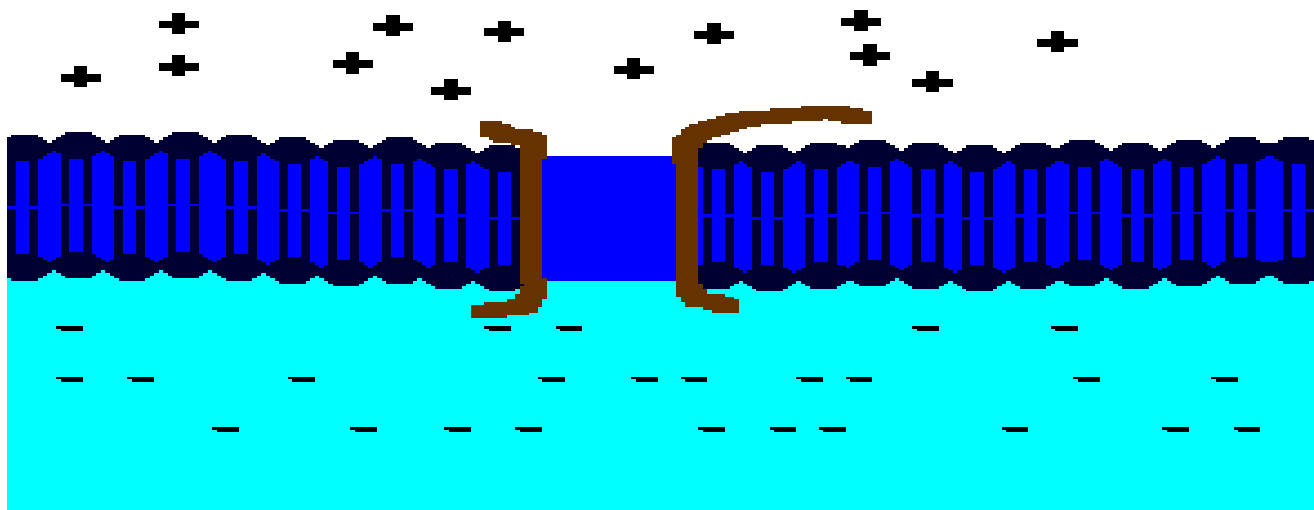
Excitation

- All cells have a **voltage difference** across their PM. i.e., inner portion of the PM is more negative than the outer. This is the result of several things:
1. ECF is very high in Na^+ while the ICF is high in K^+ . The PM is impermeable to Na^+ but slightly permeable to K^+ . As a result, K^+ is constantly leaking out of the cell. In other words, positive charge is constantly leaking out of the cell.
 2. **Na^+/K^+ pump** is constantly pumping 3 Na^+ ions **out of the cell** and 2 K^+ ions **in to the cell**. Thus more positive charge is leaving than entering.
 3. Protein anions (i.e., **negatively charged proteins**) within the ICF that cannot travel through the PM.



❑ An immediate change of the RMP into depolarization which is followed by repolarization [reestablishment of RMP] is called **action potential or nerve impulse**

■ stimulus changes the RMP to threshold level → Voltage gated Na^+ channels open → Na^+ influx → **depolarization of the membrane potential** → Voltage gated Na^+ channels closed at the same time the voltage gated K^+ channels begin to open → K^+ efflux → **repolarization of the membrane potential**

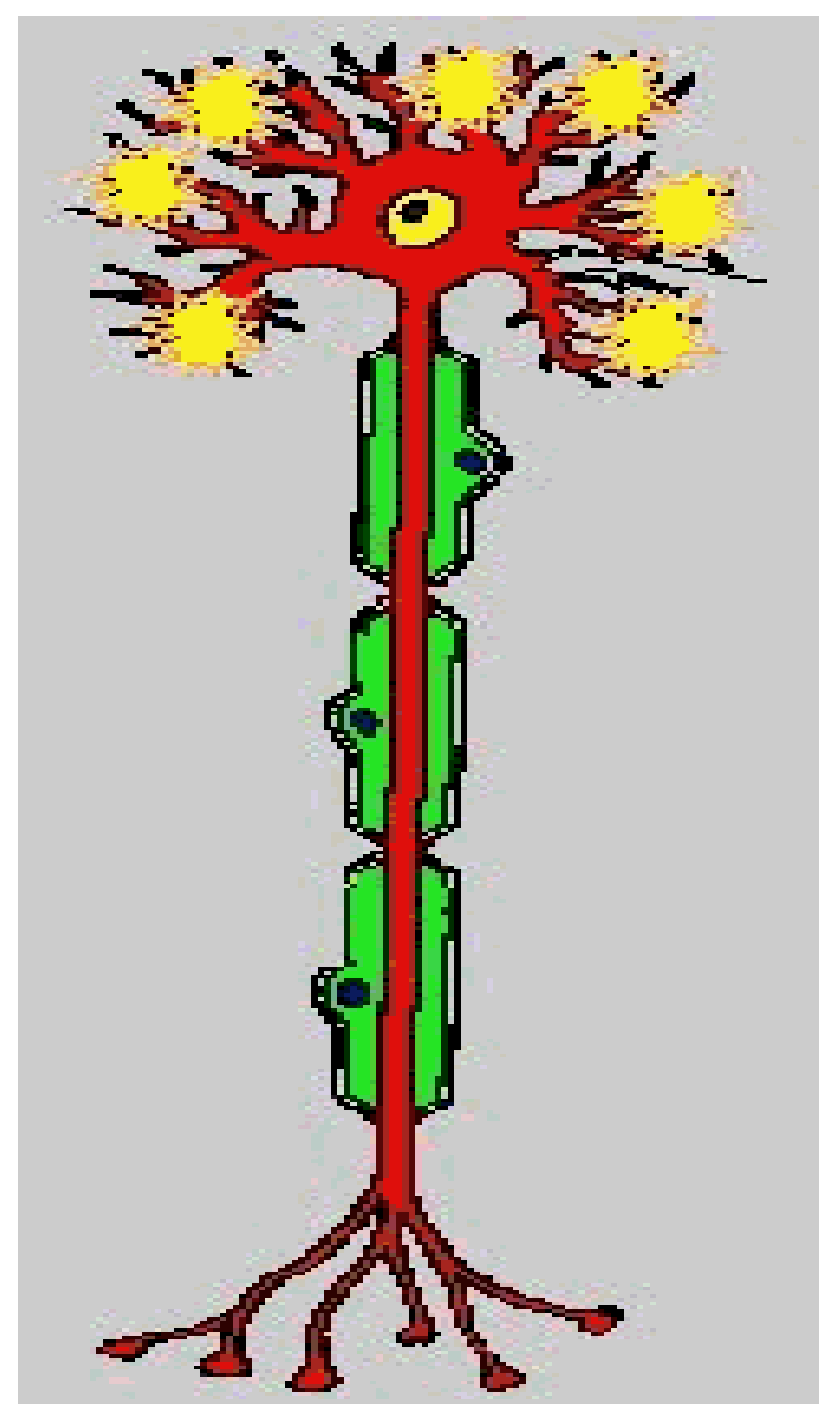
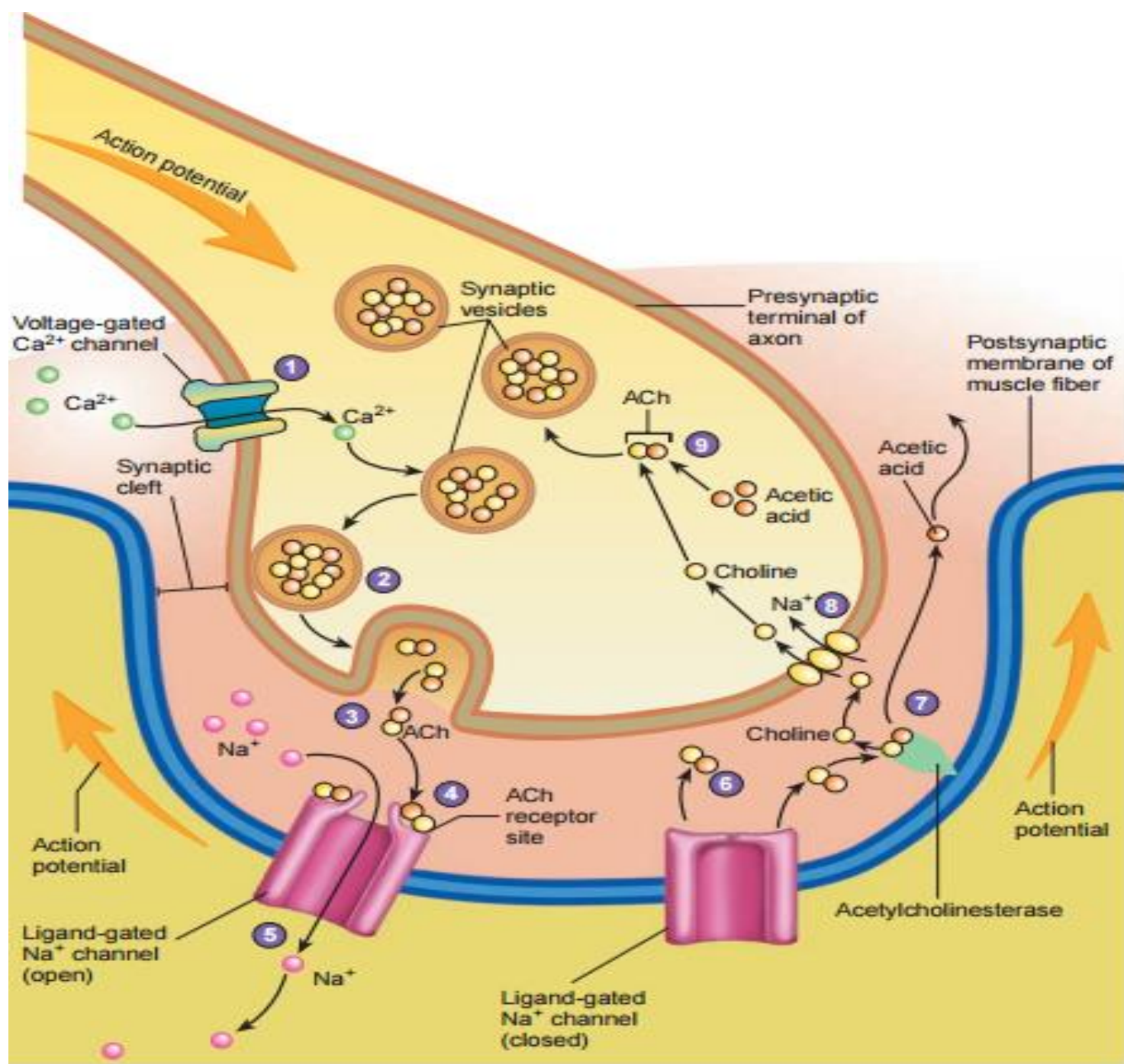


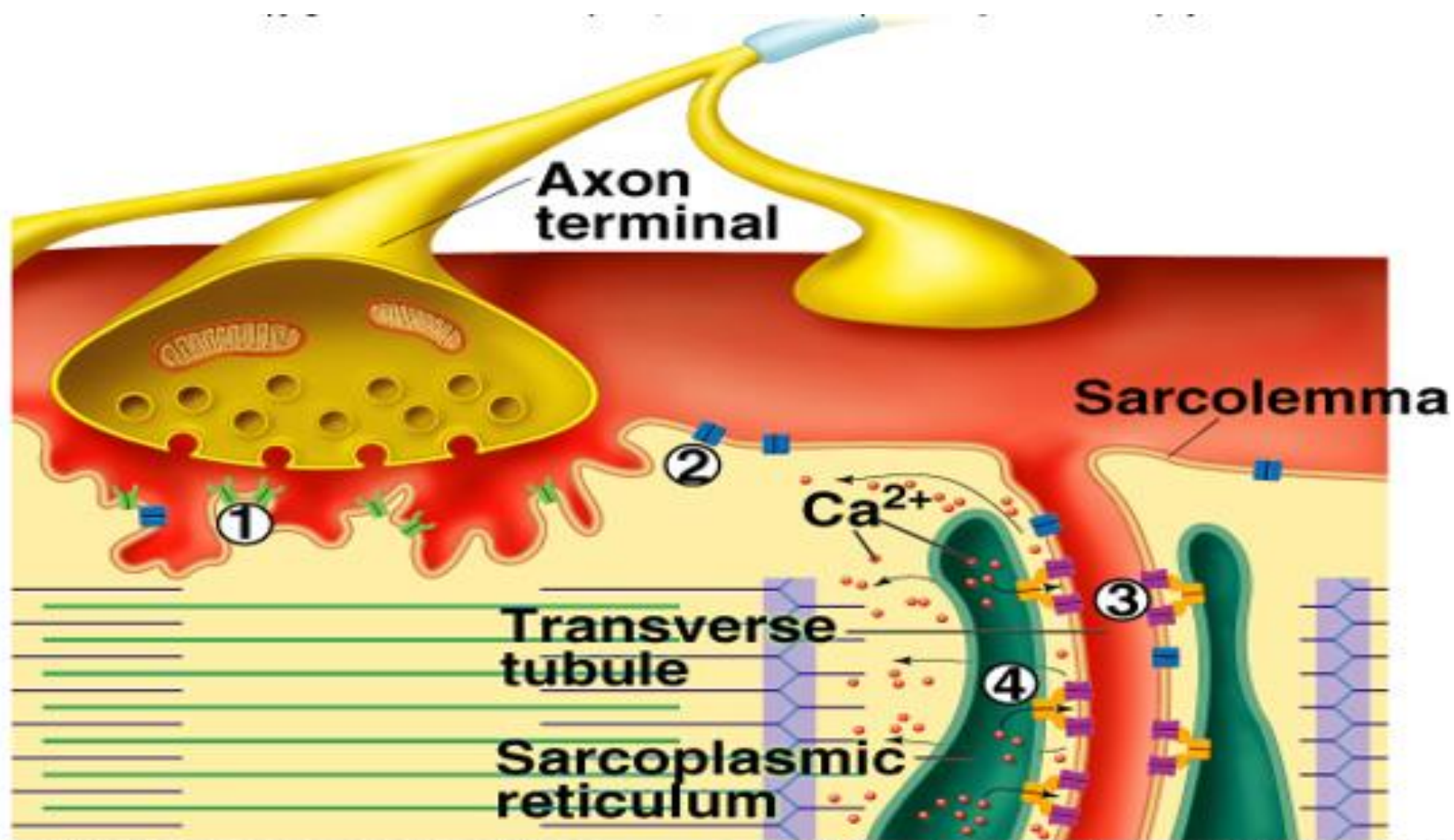
Mechanism of Muscle Contraction (Excitation-Contraction Coupling)

- when a muscle membrane is depolarized, contraction of muscle fibre follows
 - ✗ The process of depolarization initiates the contraction of muscle fibre is called **Excitation Contraction Coupling**.

□ It has several steps

1. Action potential initiated and propagated along the motor nerve fibre and arrives at the axon terminals (synaptic end bulb).
2. Opening of VG- Ca^{2+} -channels and influx of Ca^{2+} to trigger the release of Ach
3. Ach released and diffuse through the synaptic cleft and binds to NR on postjunctional membrane.
4. Opening of **ligand gated Na^{+} channels** and influx of Na^{+} to produce **postsynaptic potentials (EPSP)**.
5. Spread of depolarization through the sarcolemma
6. Spread of depolarization through the T-tubules





- ① Nicotinic acetylcholine receptor
- ② Skeletal muscle voltage-gated sodium channel

- ③ Transverse tubule voltage-gated calcium channel
- ④ Sarcoplasmic reticulum calcium release channel

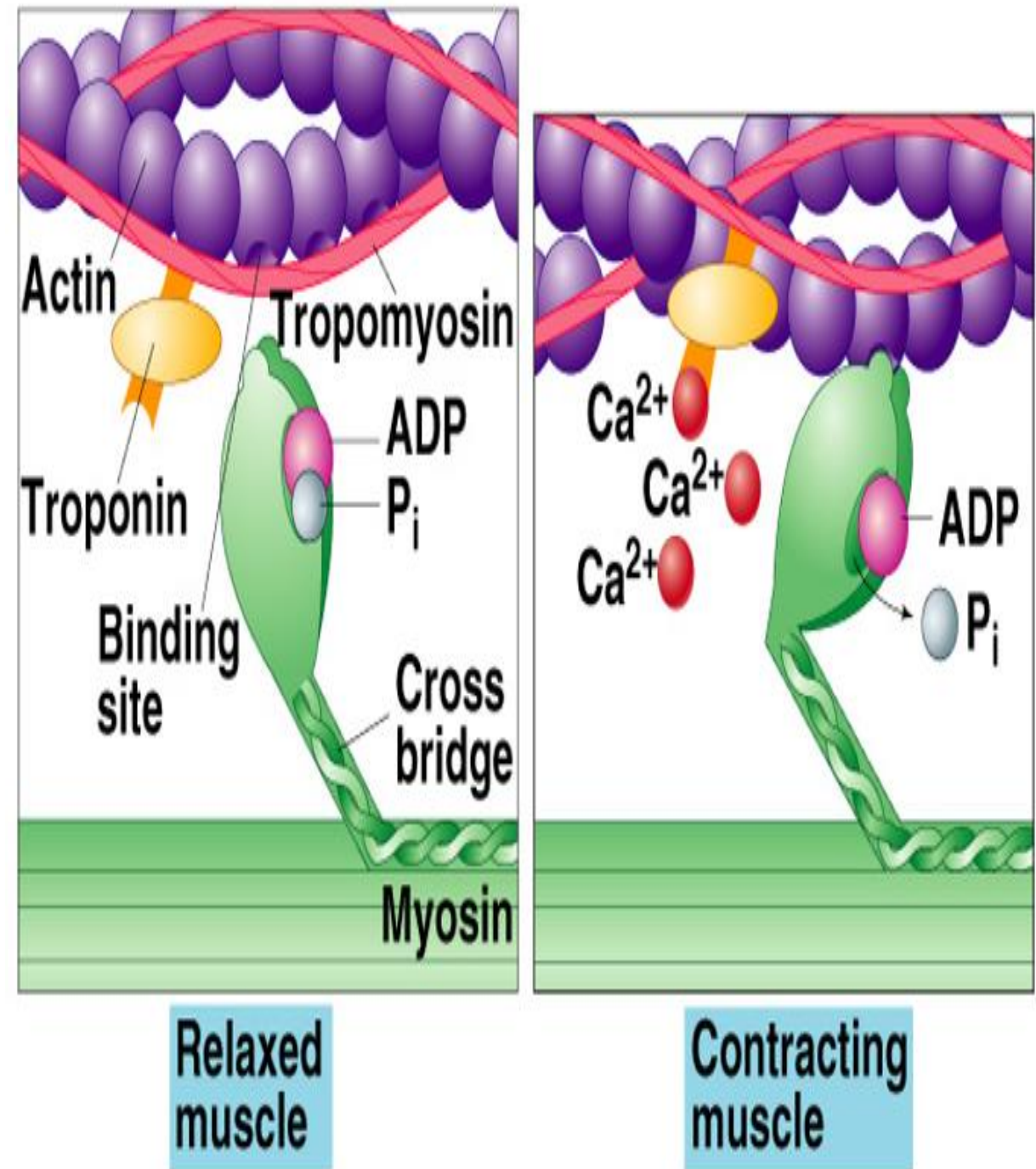
7. Depolarization of T-tubules stimulate **SR** to release Ca^{2+} into sarcoplasm
8. **Ca^{2+} binds to troponin-C**
9. Ca^{2+} and troponin-C combination **detaches troponin-I from the actin**
10. The detachment of troponin-I from the actin **displaces the tropomyocin**, uncovering the active sites of actin filaments.
11. When the active site of actin exposed, the **heads of myosin** connect with **actin**
 $\Rightarrow$ **making cross-bridges b/n myosin and actin.**
12. **ATPase enzyme** on the myosin heads hydrolyze **ATP** into **ADP + -P plus energy.**
13. The **released energy** causes the **mov't of the head** (power stroke) towards the centre $\Rightarrow$ **Muscle Contraction**
14. The **head of myosin is charged with a new molecule of ATP** and then detached from actin $\Rightarrow$ **Muscle relaxation**

Mechanism of Muscle Relaxation

❑ It has the following steps

1. Following muscle contraction, Ca^{2+} is re-uptaken back into SR by **Ca-pump**, this requires ATP.
2. Decreased Ca^{2+} in the sarcoplasm $\rightarrow$ Ca^{2+} detaches from troponin-C $\rightarrow$ Tropomyosin covers the active sites of actin.
3. Head of myosin charged with ATP, and detached from actin.

❑ Therefore, Muscle Relaxation is an active process requiring energy.





THE END